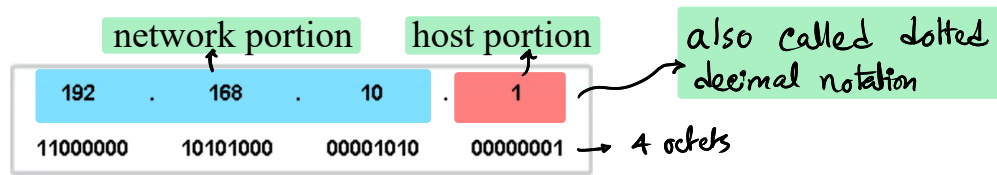


# Lecture 7 (IP Addressing)



example of an IP Address

## Ways to identify Network and Host portions:

### 1. Prefix Mask:

192.168.0.101 / 24 (meaning the first 24 out of 32 bits are network portion)

### 2. Subnet Mask:

11111111 11111111 11111111 00000000 bin  
255 . 255 . 255 . 0 dec (another way to write prefix mask uses AND operation to find network IP)

(1 for all network bits, 0 for all host bits)

## Types of IPV4 Addresses:

(ex: 192.168.1.101)

1. **Network Address:** address of the whole network

192.168.1.0 (host value all bits 0)

2. **Broadcast Address:** address to broadcast to all devices in the network

192.168.1.255 (host value all bits 1, usually 255) *→ given that host bits = 8*

3. **Host Address:** address of individual devices

192.168.1.112 (host value between 1 and 254) *→ given that host bits = 8*

## Types of Casting:

1. **Unicast:** message to one host.

destination address: host's IP Address

2. **Broadcast:** message to all hosts in a network

**Limited Broadcast:** broadcast to own network.

destination address: 255.255.255.255

Routers are not involved

**Directed Broadcast:** broadcast to another network

destination address: broadcast address of target network

3. **Multicast:** message to a specific group of hosts

destination address: uses a separate multicast address ranging from 224-239

Honorable mention:

**Loopback Address:** address for messages from the device back to itself

format: 127.x.x.x/8 *→ 8 bits Fixed (127)*

*↑ must start with 127, then any address in between*

Identify the type of address

1. 192.168.10.255/23
2. 192.168.11.0/23
3. 192.168.10.128/25
4. 192.168.10.127/25

broadcast

network

host

host

initial assumption

Breakdown: 1.  $\frac{192}{8} \cdot \frac{168}{8} \cdot \frac{10}{7} \cdot \frac{255}{7} / 23$  → network address bits

00001010 11111111  
10 255

host value not all 1, hence this is

not broadcast, this is address

host address

For the following IP address 1.2.168.10/20 Find

① Since network bits = 20  
∴ host bits = 32 - 20 = 12

∴ Total usable addresses:

$$2^{12} - 2 = 4094$$

↙      ↘  
network address    broadcast address

② Since network bits = 20

Subnet mask:

11111111 11111111 11110000 00000000  
255 . 255 . 240 . 0

① Total usable IP addresses

② Subnet Mask

③ Network Address

④ Broadcast address 1.2.175.255/20

⑤ 2nd Usable address

⑥ 2nd last usable address

③ Given IP:

1.2.168.10/20

$\frac{1}{8} \cdot \frac{2}{8} \cdot \frac{16}{4} \cdot \frac{10}{8}$   
10100000 00000000  
1 . 2 . 160 . 0

Network Address: 1.2.160.0/20

#### IV Broadcast address:

1.2.168.10/20

↓  
0000 0001 . 0000 0010 . 1010 1000 . 0000 1010  
↓  
0000 0001 . 0000 0010 . 1010 1111 . 1111 1111  
↓  
1.2.175.255

1.2.160.0 → Network Address

• 1 → 1st usable

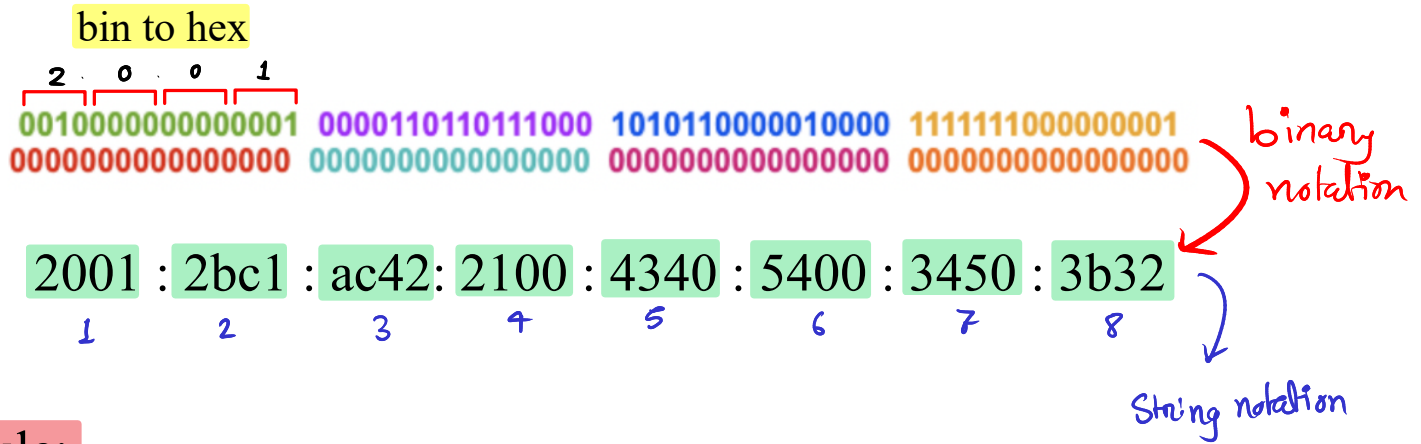
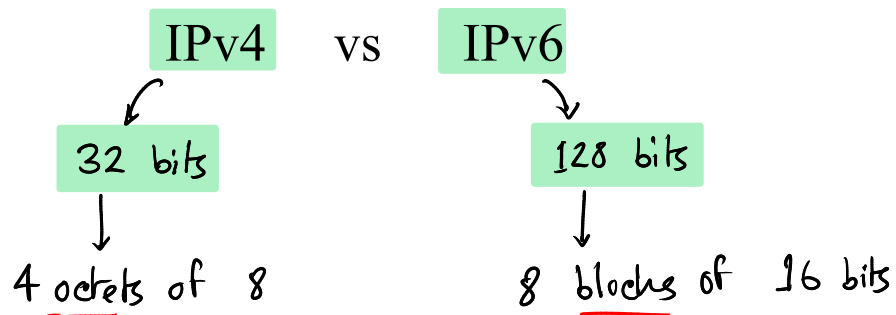
• 2 → 2nd usable

• 253 → 2nd last usable

• 254 → last usable

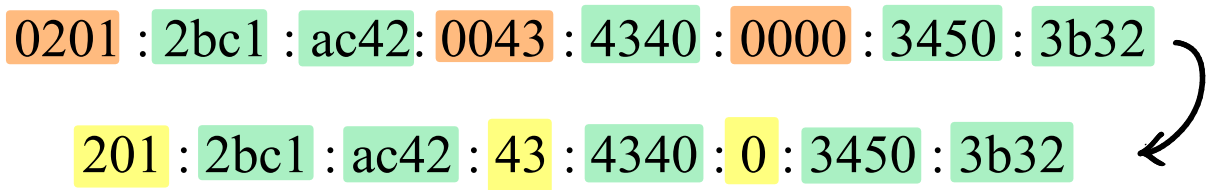
1.2.175.255 → Broadcast Address

# IPv6:



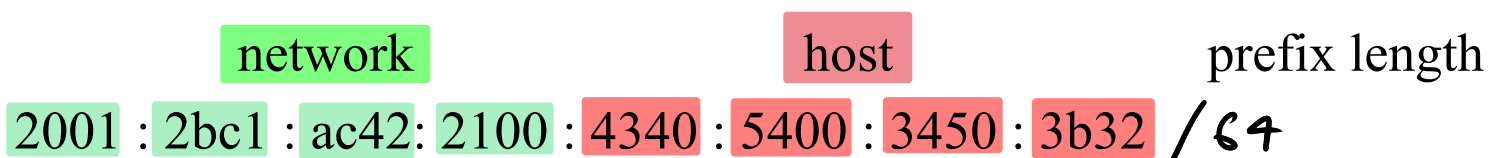
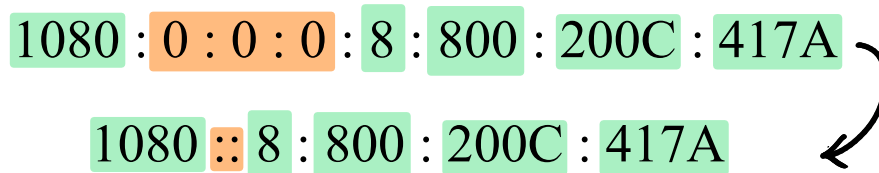
## 1st rule:

leading zeros don't need to be written



## 2nd rule:

continuous strings of 0's can be substituted with ::, but can be only used once in the whole address due to ambiguity issues



# Lecture 8: (Subnetting)

Solutions to IPv4 Address shortage:

1. Shifting to Ipv6 } long term solution

2. Subnetting

3. Private Addressing } short term solution

**Subnetting** is partitioning a single network into smaller group of networks (subnets) by borrowing bits from the host part to the network part

done in two ways:

- i. Fixed Length Subnet Masking → prefix length equal for each network
- ii. Variable Length Subnet Masking → network with more hosts gets more host addresses

## Fixed Length Subnet Masking:

lets say we have 4 networks and we are given the following network address: **200.20.20.0 / 24**

From the following network,

$$\text{host bit} = 32 - 24 = 8$$

already a network address, otherwise needs conversion

∴ Since 4 networks, we need 2 bits from host portion:

|    |               |                                        |                             |
|----|---------------|----------------------------------------|-----------------------------|
| 00 | 00 0000 → 0   | ∴ network address = 200.20.20.0 / 26   | (Since 2 bits were shifted) |
| 01 | 00 0000 → 64  | ∴ network address = 200.20.20.64 / 26  |                             |
| 10 | 00 0000 → 128 | ∴ network address = 200.20.20.128 / 26 |                             |
| 11 | 00 0000 → 192 | ∴ network address = 200.20.20.192 / 26 |                             |

← shifted to network portion

∴ depending on the number of networks we need to shift bits so that the new possible combinations from the shifted bits can assign a network address for each of the network

for 2 networks → 2 bits → 4 combinations

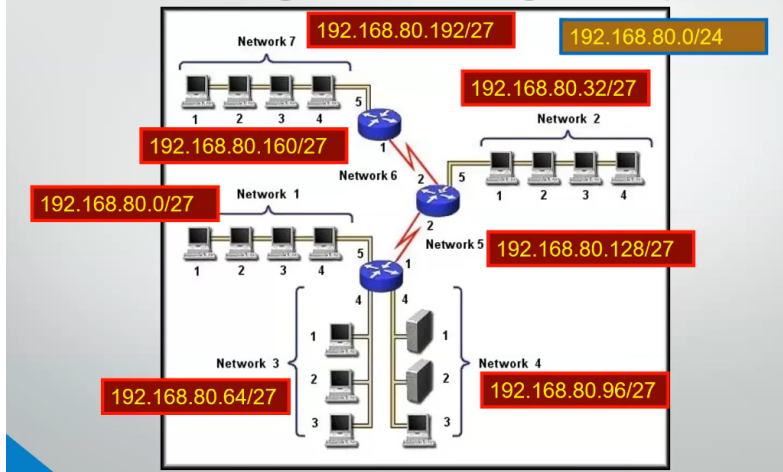
for 9 networks → 4 bits → 16 combinations

for 21 networks → 5 bits → 32 combinations

for 30 networks → 5 bits → 32 combinations

for 16 networks → 4 bits → 16 combinations

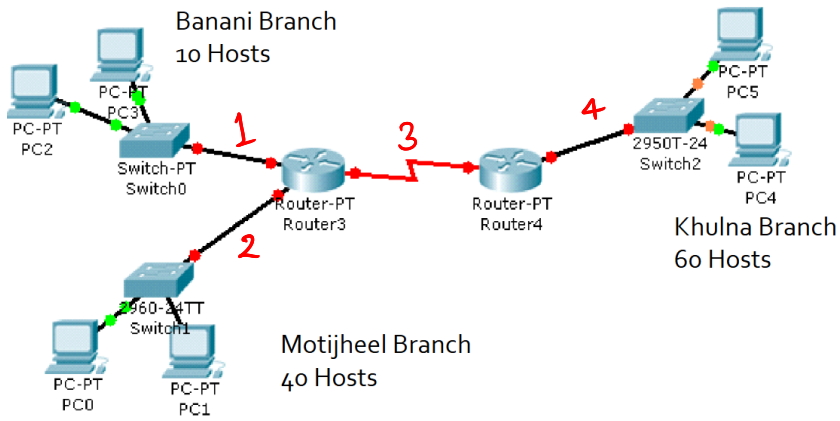
## Fixed Length Subnetting: Example



192.168.80.0/24

Since 7 networks, 3 bits needed!

|       |           |                   |
|-------|-----------|-------------------|
| 0 0 0 | 0 0 0 0 0 | 192.168.80.0/27   |
| 0 0 1 | 0 0 0 0 0 | 192.168.80.32/27  |
| 0 1 0 | 0 0 0 0 0 | 192.168.80.64/27  |
| 0 1 1 | 0 0 0 0 0 | 192.168.80.96/27  |
| 1 0 0 | 0 0 0 0 0 | 192.168.80.128/27 |
| 1 0 1 | 0 0 0 0 0 | 192.168.80.160/27 |
| 1 1 0 | 0 0 0 0 0 | 192.168.80.192/27 |

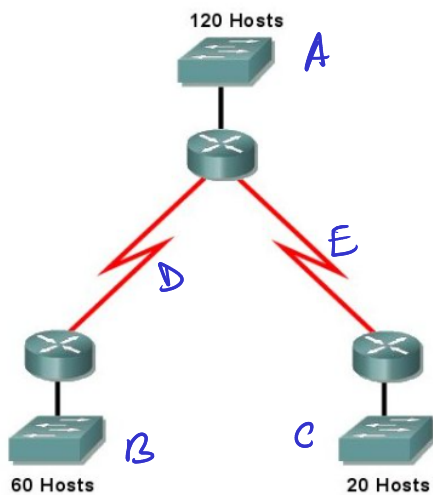


the following network if allocated with fixed length subnetting will get 32-26 or 6 host bits for each network, hence  $2^6 = 64$  IP addresses, but network 1, 2, 3, 4 needs 3,3,2,3 IP Addresses, so we can see using Fixed Length Subnetting wastes a lot of IP address.

the solution:

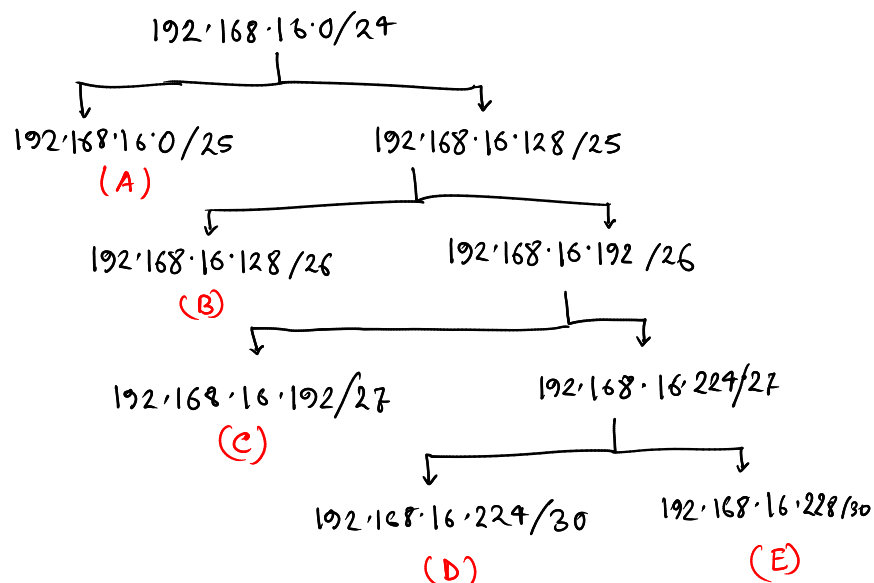
## Variable Length Subnetting:

192.168.16.0 / 24



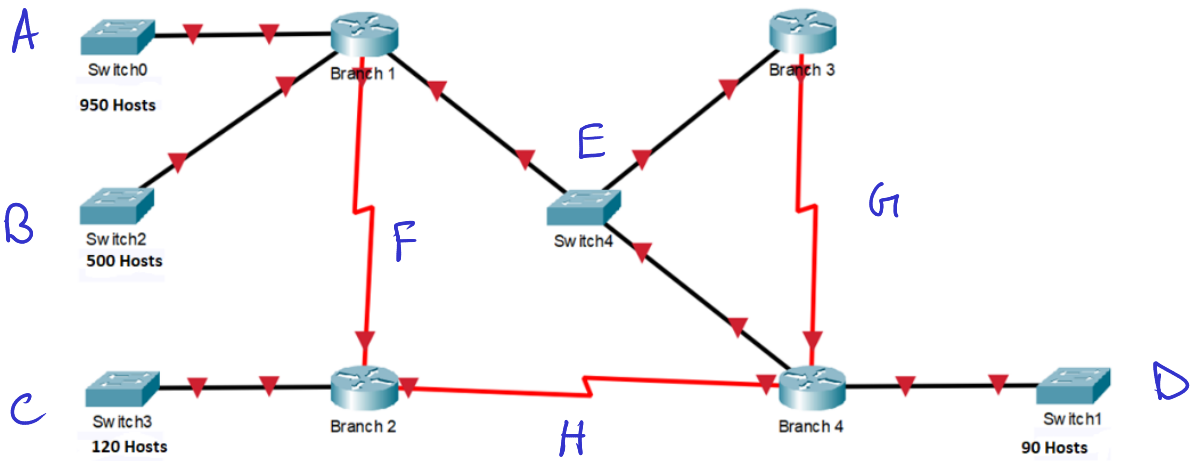
| Name | Host | +2  | Bits | Prefix | Waste |
|------|------|-----|------|--------|-------|
| A    | 120  | 122 | 7    | 25     | 6     |
| B    | 60   | 62  | 6    | 26     | 4     |
| C    | 20   | 22  | 5    | 27     | 10    |
| D    | 2    | 4   | 2    | 30     | 0     |
| E    | 2    | 4   | 2    | 30     | 0     |

0 | 0 0 0 0 0 0 0 0 → 0 → 192.168.16.0 / 25 (A)  
 1 | 0 0 0 0 0 0 0 0 → 128 → 192.168.16.128 / 26 (B)  
 1 | 1 0 0 0 0 0 0 0 → 192 → 192.168.16.192 / 27 (C)  
 1 | 1 1 0 0 0 0 0 0 → 224 → 192.168.16.224 / 30 (D)  
 1 | 1 1 0 0 1 0 0 0 → 232 → 192.168.16.228 / 30 (E)





Given Network Address : 13.65.136.0/21



| Name | Hosts | +2  | Host Bits | Prefix | Waste |
|------|-------|-----|-----------|--------|-------|
| A    | 950   | 952 | 10        | 22     | 72    |
| B    | 500   | 502 | 9         | 23     | 10    |
| C    | 120   | 122 | 7         | 25     | 6     |
| D    | 90    | 92  | 7         | 25     | 36    |
| E    | 3     | 5   | 3         | 29     | 3     |
| F    | 2     | 4   | 2         | 30     | 0     |
| G    | 2     | 4   | 2         | 30     | 0     |
| H    | 2     | 4   | 2         | 30     | 0     |

given IP: 13.65.136.0/21

$100010000000000000000000 \rightarrow 136.0 \rightarrow 13.65.136.0/22$  (A)  
 $100011000000000000000000 \rightarrow 140.0 \rightarrow 13.65.140.0/23$  (B)  
 $100011100000000000000000 \rightarrow 142.0 \rightarrow 13.65.142.0/25$  (C)  
 $100011100000000000000000 \rightarrow 142.128 \rightarrow 13.65.142.128/25$  (D)  
 $100011100000000000000000 \rightarrow 143.0 \rightarrow 13.65.143.0/29$   
 $100011100000000000000000 \rightarrow 143.16 \rightarrow 13.65.143.8/30$   
 $100011100000000000000000 \rightarrow 143.20 \rightarrow 13.65.143.12/30$   
 $100011100000000000000000 \rightarrow 143.24 \rightarrow 13.65.143.16/30$

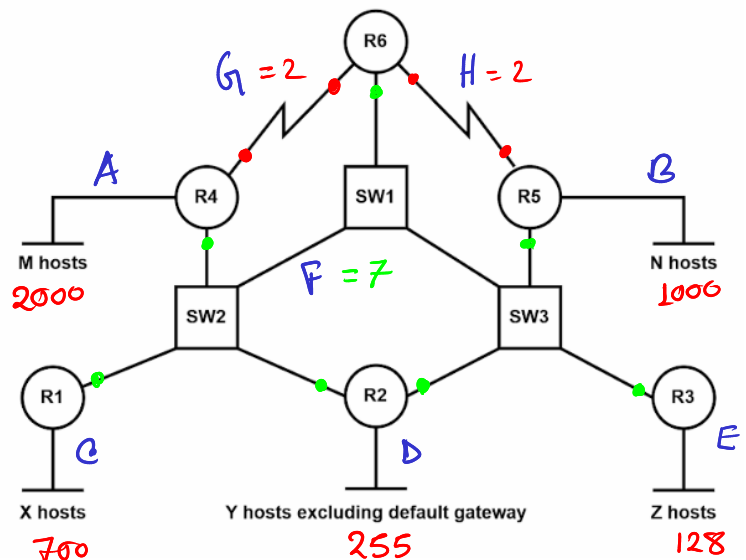
2 4 8 16 32 64 128 256 512 1024 2048  
1 2 3 4 5 6 7 8 9 10 11

| Name | Host | Host+2 | Block size | Host bits | Prefix | Waste |
|------|------|--------|------------|-----------|--------|-------|
| A    | 2000 | 2002   | 2048       | 11        | 21     | 46    |
| B    | 1000 | 1002   | 1024       | 10        | 22     | 22    |
| C    | 700  | 702    | 1024       | 10        | 22     | 322   |
| D    | 255  | 257    | 512        | 9         | 23     | 255   |
| E    | 128  | 130    | 256        | 8         | 24     | 126   |
| F    | 7    | 9      | 16         | 4         | 28     | 7     |
| G    | 2    | 4      | 4          | 2         | 30     | 0     |
| H    | 2    | 4      | 4          | 2         | 30     | 0     |

x y

y-x  
778

Given the network address is 1.17.128.0/17, the value of M=2000, N=1000, X=700, Y=254, Z=128, answer the following questions.



Given network: 1.17.128.0/17

∴ host bits: 1|000 0|000 0000 0000 → 128.0 (A)  
 1000 1|000 0000 0000 → 136.0 (B)  
 1000 11|000 0000 0000 → 140.0 (C)  
 1001 000|0 0000 0000 → 144.0 (D)  
 1001 001|0 0000 0000 → 146.0 (E)  
 1001 0011|0 0000 0000 → 147.0 (F)  
 1001 0011 0000|0 0000 → 147.16 (G)  
 1001 0011 0001|0 0000 → 147.20 (H)  
 1001 0011 0001 01|00

1.17.128.0/17

